

第 10 周习题课参考内容

二重积分的计算和性质

一、计算二重积分

1. $\iint_D x \cos(xy) d\sigma$, $D = [0, 2\pi] \times [0, 1]$ 【练习题 16.3: 1(2)】

$$\text{解: } \iint_D x \cos(xy) d\sigma = \int_0^{\pi/2} dx \int_0^1 x \cos(xy) dy = \int_0^{\pi/2} [\sin(xy)]_{y=0}^{y=1} dx = \int_0^{\pi/2} \sin x dx = 1。$$

注: 观察另一种积分次序 $\iint_D x \cos(xy) d\sigma = \int_0^1 dy \int_0^{\pi/2} x \cos(xy) dx = \dots$ (积分较麻烦)

2. $\iint_D \sin(x+y) d\sigma$, $D = [0, \pi] \times [0, 1]$ 【练习题 14.3: 1(3)类似】

$$\begin{aligned} \text{解: } \iint_D \sin(x+y) d\sigma &= \int_0^{\pi} dx \int_0^1 \sin(x+y) dy = \int_0^{\pi} [-\cos(x+y)]_{y=0}^{y=1} dx \\ &= \int_0^{\pi} [\cos x - \cos(x+1)] dx = [\sin x - \sin(x+1)]_0^{\pi} = \sin 1 - \sin(\pi+1)。 \end{aligned}$$

注: 换一个积分次序 $\iint_D \sin(x+y) d\sigma = \int_0^{\pi} dy \int_0^1 \sin(x+y) dx = \dots$ (难度一样)

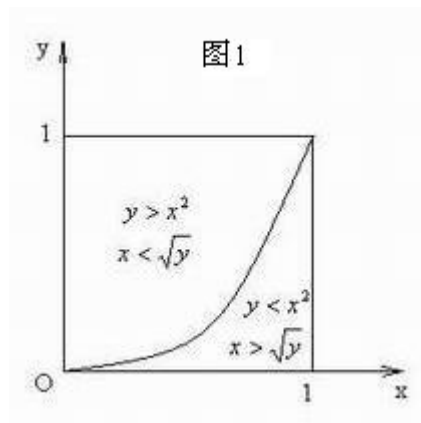
3. $\iint_D f(x, y) d\sigma$, $D = [0, 1] \times [0, 1]$, 【练习题 14.3: 2】

$$\text{其中 (1) } f(x, y) = \begin{cases} 1, & y \leq x^2, \\ 0, & y > x^2; \end{cases} \quad (2) \quad f(x, y) = \begin{cases} x+y, & x^2 \leq y \leq 2x^2, \\ 0, & y < x^2 \text{ 或 } y > 2x^2. \end{cases}$$

解: (1) 参考图 1, 可以采用两种积分次序

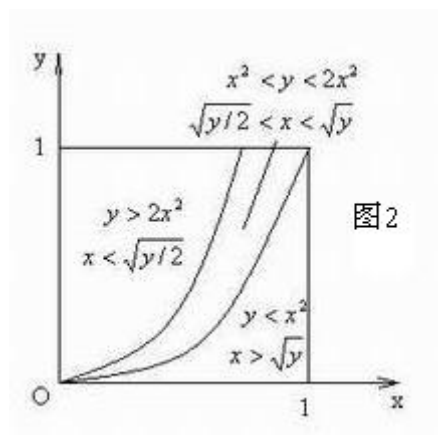
$$\begin{aligned} \iint_D f(x, y) d\sigma &= \int_0^1 dx \int_0^{x^2} 1 dy \\ &= \int_0^1 x^2 dx = \frac{1}{3}; \\ \iint_D f(x, y) d\sigma &= \int_0^1 dy \int_{\sqrt{y}}^1 1 dx \\ &= \int_0^1 (1 - \sqrt{y}) dy = \frac{1}{3}。 \end{aligned}$$

(两种算法结果相符)



(2) 参考图 2, 采用下面的积分次序比较容易

$$\begin{aligned}
 \iint_D f(x,y) d\sigma &= \int_0^1 dy \int_{\sqrt{y/2}}^{\sqrt{y}} (x+y) dx \\
 &= \int_0^1 \left(\frac{x^2}{2} + yx \right) \Big|_{\sqrt{y/2}}^{\sqrt{y}} dy = \int_0^1 \left[\frac{1}{4}y + \left(1 - \frac{1}{\sqrt{2}}\right)y^{3/2} \right] dy \\
 &= \frac{1}{8} + \left(1 - \frac{1}{\sqrt{2}}\right) \frac{2}{5} = \frac{21}{40} - \frac{\sqrt{2}}{5}.
 \end{aligned}$$



思考：换一种次序 $\iint_D f(x,y) dA = \int_0^1 dx \int_{x^2}^{2x^2} (x+y) dy$ 为什么不正确？

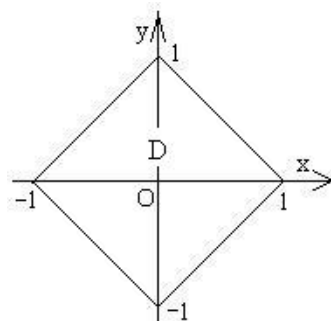
二、将二重积分化为累次积分计算

【选自练习题 16.5：1，化为累次积分，不必具体计算】

1. $\iint_D e^{x+y} d\sigma$, $D = \{(x,y) : |x| + |y| \leq 1\}$

解：D 由 $x \pm y = \pm 1$ （4 条直线）围成，如右图，

$$\iint_D e^{x+y} dx dy = \int_{-1}^0 dx \int_{-x-1}^{x+1} e^{x+y} dy + \int_0^1 dx \int_{x-1}^{1-x} e^{x+y} dy = \dots$$



2. $\iint_D x \cos(xy) d\sigma$, $D = \{(x,y) : x^2 + y^2 \leq a^2\}$

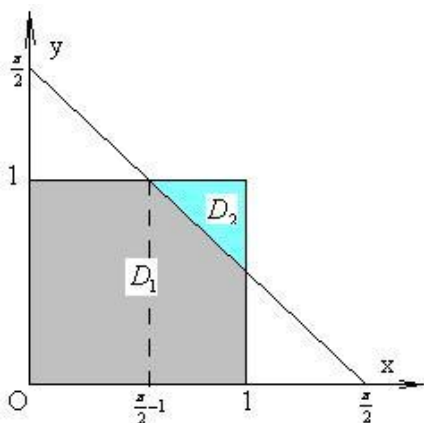
解：D 由 $y = \pm \sqrt{a^2 - x^2}$ ($|x| \leq a$) 围成，还可以利用区域和函数的对称性，

$$\iint_D x \cos(xy) d\sigma = \int_{-a}^a dx \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} x \cos(xy) dy = \dots$$

3. $\iint_D |\cos(x+y)| d\sigma$, $D = [0,1] \times [0,1]$

解：根据 $\cos$ 函数的符号，用直线 $x+y = \pi/2$

将区域 D 分成两个子区域 $D = D_1 \cup D_2$ ，如左图，



$$\begin{aligned}
 \text{则 } \iint_D &= \iint_{D_1} + \iint_{D_2} \\
 \iint_{D_1} &= \int_0^{\frac{\pi}{2}-1} dx \int_0^1 \cos(x+y) dy + \int_{\frac{\pi}{2}-1}^1 dx \int_0^{\frac{\pi}{2}-x} \cos(x+y) dy \\
 &= \dots
 \end{aligned}$$

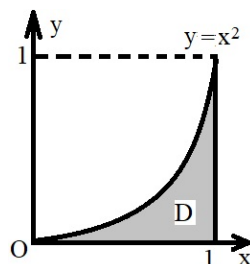
$$\iint_{D_2} = - \int_{\frac{\pi}{2}-1}^1 dx \int_{\frac{\pi}{2}-x}^1 \cos(x+y) dy = \dots$$

三、改变累次积分次序【练习题 16.5: 2】

(利用二重积分过渡, 作出平面区域图形来说明)

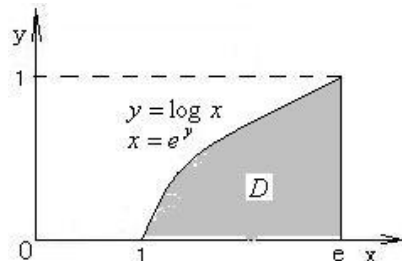
$$1. \int_0^1 dx \int_0^{x^2} f(x, y) dy$$

解: 如右图所示, 原式 = $\iint_D f(x, y) d\sigma = \int_0^1 dy \int_{\sqrt{y}}^1 f(x, y) dx$



$$2. \int_1^e dx \int_0^{\ln x} f(x, y) dy$$

解: 如右图, 原式 = $\iint_D f(x, y) d\sigma = \int_0^1 dy \int_{e^y}^e f(x, y) dx$

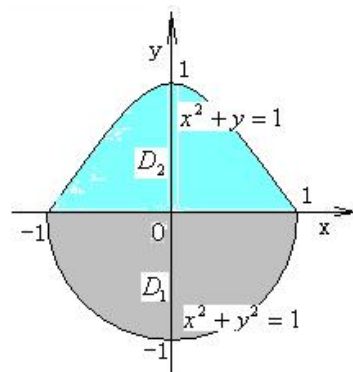


$$3. \int_0^1 dx \int_{x^2}^x f(x, y) dy$$

解: 自己作图, 原式 = $\iint_D f(x, y) d\sigma = \int_0^1 dy \int_y^{\sqrt{y}} f(x, y) dx$

$$4. \int_a^b dx \int_a^x f(x, y) dy$$

解: 自己作图, 原式 = $\iint_D f(x, y) d\sigma = \int_a^b dy \int_y^b f(x, y) dx$



$$5. \int_{-1}^1 dx \int_{-\sqrt{1-x^2}}^{1-x^2} f(x, y) dy$$

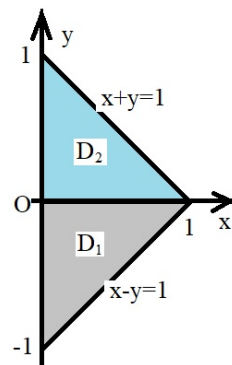
解: 如右上图,

$$\text{原式} = \iint_{D_1} + \iint_{D_2} = \int_{-1}^0 dy \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f(x, y) dx + \int_0^1 dy \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f(x, y) dx$$

$$6. \int_0^1 dx \int_{x-1}^{1-x} f(x, y) dy$$

解: 如右图所示,

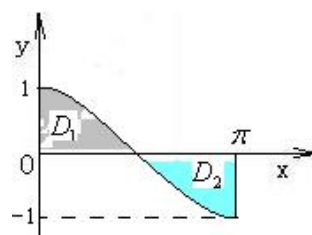
$$\text{原式} = \iint_{D_1} + \iint_{D_2} = \int_{-1}^0 dy \int_0^{1+y} f(x, y) dx + \int_0^1 dy \int_0^{1-y} f(x, y) dx$$



$$7. \int_0^{\pi} dx \int_0^{\cos x} f(x, y) dy$$

解: 如右图, 注意在 D_2 中积分需要增加负号

$$\text{原式} = \iint_{D_1} - \iint_{D_2} = \int_0^1 dy \int_0^{\arccos y} f(x, y) dx - \int_{-1}^0 dy \int_{\arccos y}^{\pi} f(x, y) dx$$



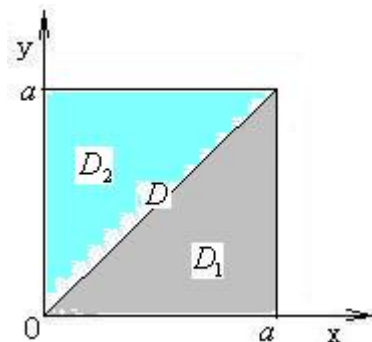
四、综合练习

1. 设 f 为一元连续函数, 求证 $\int_0^a dx \int_0^x f(x)f(y)dy = \frac{1}{2} \left[\int_0^a f(x)dx \right]^2$ 【练习题 16.5: 3】

证明: 如右图, 左式可化为区域 D_1 上的二重积分,

再利用区域和被积函数关于 x 和 y 的对称性, 以及积分的区域可加性, 便有

$$\begin{aligned} \text{左式} &= \iint_{D_1} f(x)f(y)d\sigma = \iint_{D_2} f(x)f(y)d\sigma \\ &= \frac{1}{2} \iint_D f(x)f(y)d\sigma = \frac{1}{2} \int_0^a dx \int_0^a f(x)f(y)dy \\ &= \frac{1}{2} \int_0^a f(x)dx \int_0^a f(y)dy = \text{右式} \quad \square \end{aligned}$$



2. 设 $f(x)$ 在 $[a, b]$ 上连续, 利用二重积分证明:

$$\left[\int_a^b f(x)dx \right]^2 \leq (b-a) \int_a^b f^2(x)dx$$

其中等号当且仅当 $f(x)$ 为常数时成立。

证明: 已知 $f(x)$ 在 $[a, b]$ 上连续, 考虑

$$D = \{(x, y) | a \leq x \leq b, a \leq y \leq b\} \text{ 上的 2 元连续函数 } [f(x) - f(y)]^2 \geq 0,$$

恒等号成立当且仅当 $f(x)$ 为常数; 由此

$$\begin{aligned} 0 &\leq \int_a^b dx \int_a^b (f(x) - f(y))^2 dy \\ &= \int_a^b dx \int_a^b f^2(x) dy + \int_a^b dx \int_a^b f^2(y) dy - 2 \int_a^b dx \int_a^b f(x)f(y) dy \\ &= (b-a) \int_a^b f^2(x) dx + (b-a) \int_a^b f^2(y) dy - 2 \left[\int_a^b f(x) dx \right] \left[\int_a^b f(y) dy \right] \\ &= 2(b-a) \int_a^b f^2(x) dx - 2 \left[\int_a^b f(x) dx \right]^2 \end{aligned}$$

$$\text{因此} \quad \left[\int_a^b f(x) dx \right]^2 \leq (b-a) \int_a^b f^2(x) dx,$$

其中等号当且仅当 $f(x)$ 为常数时成立。 $\square$

3. 不计算, 判断二重积分 $\iint_{x^2+y^2 \leq 4} \sqrt[3]{1-x^2-y^2} d\sigma$ 的符号。

解: 将积分区域分解 $D = D_1 \cup D_2 \cup D_3$, 其中 $D_1 = \{(x, y) | x^2 + y^2 \leq 1\}$,

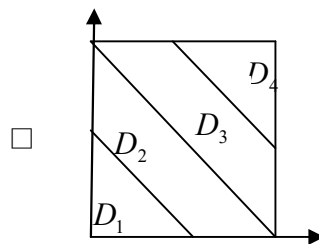
$$\begin{aligned}
 D_2 &= \{(x, y) | 1 \leq x^2 + y^2 \leq 2\}, \quad D_3 = \{(x, y) | 2 \leq x^2 + y^2 \leq 4\} \\
 \text{则} \quad \iint_D \sqrt[3]{1-x^2-y^2} d\sigma &= \iint_{D_1} + \iint_{D_2} + \iint_{D_3}, \\
 \iint_{D_1} \sqrt[3]{1-x^2-y^2} d\sigma &\leq \iint_{D_1} 1 \cdot d\sigma = \sigma(D_1) = \pi, \\
 \iint_{D_2} \sqrt[3]{1-x^2-y^2} d\sigma &\leq 0, \\
 \iint_{D_3} \sqrt[3]{1-x^2-y^2} d\sigma &\leq \iint_{D_3} -1 \cdot d\sigma = -\sigma(D_3) = -2\pi, \\
 \text{故} \quad \iint_{x^2+y^2 \leq 4} \sqrt[3]{1-x^2-y^2} d\sigma &= \iint_{D_1} + \iint_{D_2} + \iint_{D_3} \leq -\pi < 0. \quad \square
 \end{aligned}$$

4. 求 $I = \iint_D [x+y] d\sigma$, 其中 $D = [0, 2] \times [0, 2]$, $[x+y]$ 为取整函数。【练习题 16.5: 1(8)】

解: 如图 $D = D_1 \cup D_2 \cup D_3 \cup D_4$, 在 D_1, D_2, D_3, D_4 中分别有

$$0 \leq x+y \leq 1, \quad 1 \leq x+y \leq 2, \quad 2 \leq x+y \leq 3, \quad 3 \leq x+y \leq 4,$$

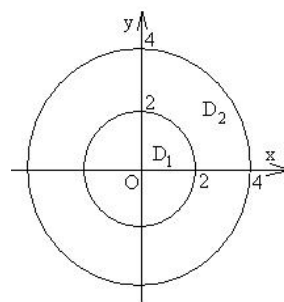
$$\begin{aligned}
 I &= \iint_D [x+y] d\sigma = \iint_{D_1} + \iint_{D_2} + \iint_{D_3} + \iint_{D_4} [x+y] d\sigma \\
 &= \iint_{D_1} 0 d\sigma + \iint_{D_2} 1 d\sigma + \iint_{D_3} 2 d\sigma + \iint_{D_4} 3 d\sigma \\
 &= 0\sigma(D_1) + 1\sigma(D_2) + 2\sigma(D_3) + 3\sigma(D_4) = \frac{3}{2}\sigma(D) = 6
 \end{aligned}$$



5. 计算 $I = \iint_D |x^2 + y^2 - 4| d\sigma$, $D: x^2 + y^2 \leq 16$ 。(本题需利用极坐标计算-周三讲解)

解: 记 $D = D_1 \cup D_2$, $D_1: x^2 + y^2 \leq 4$, $D_2: 4 \leq x^2 + y^2 \leq 16$, 则

$$\begin{aligned}
 I &= \iint_{D_1} (4 - x^2 - y^2) d\sigma + \iint_{D_2} (x^2 + y^2 - 4) d\sigma \\
 &= \int_0^{2\pi} d\phi \int_0^2 (4 - r^2) r dr + \int_0^{2\pi} d\phi \int_2^4 (r^2 - 4) r dr \\
 &= 2\pi \left(2r^2 - \frac{r^4}{4} \right) \Big|_0^2 + 2\pi \left(\frac{r^4}{4} - 2r^2 \right) \Big|_2^4 = 80\pi
 \end{aligned}$$



$$\begin{aligned}
 \text{或者} \quad I &= -\iint_{D_1} (x^2 + y^2 - 4) d\sigma + \iint_{D_2} (x^2 + y^2 - 4) d\sigma \\
 &= \iint_{D_1 \cup D_2} (x^2 + y^2 - 4) d\sigma - 2 \iint_{D_1} (x^2 + y^2 - 4) d\sigma \\
 &= \int_0^{2\pi} d\phi \int_0^4 (r^2 - 4) r dr - 2 \int_0^{2\pi} d\phi \int_0^2 (r^2 - 4) r dr = 80\pi
 \end{aligned}$$

□

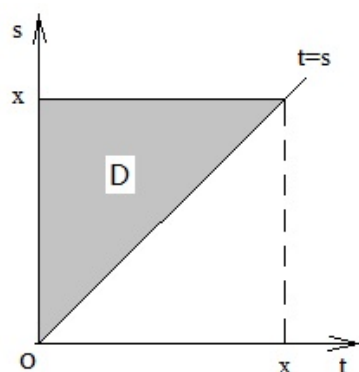
6. 设 $f(x) = \int_0^x \left[\int_t^x e^{-s^2} ds \right] dt$, 求 $f'(x)$ 与 $f(x)$ 。

解: $f(x)$ 是变上限积分, 但被积函数中又含变上限积分。

为消除被积函数中的变量 x , 考虑交换累次积分次序, 为此利用二重积分过渡如下:

记 $D = \{(s, t) | 0 \leq t \leq x, t \leq s \leq x\}$

$= \{(s, t) | 0 \leq t \leq s, 0 \leq s \leq x\}$ (见右图)



$$\begin{aligned} \text{则 } f(x) &= \int_0^x \left[\int_t^x e^{-s^2} ds \right] dt = \iint_D e^{-s^2} ds dt \\ &= \int_0^x \left[\int_0^s e^{-s^2} dt \right] ds = \int_0^x s e^{-s^2} ds = \frac{1}{2} (1 - e^{-x^2}), \\ f'(x) &= \frac{d}{dx} \int_0^x s e^{-s^2} dt = x e^{-x^2}. \end{aligned}$$

□

五、选讲/选做题*

1. 计算积分 $I = \int_0^{\frac{\pi}{2}} \ln \frac{1+a \cos x}{1-a \cos x} \frac{dx}{\cos x}$, ($|a| < 1$)

解: 直接计算比较困难, 先观察一下: 对于 $0 \leq x \leq \frac{\pi}{2}$,

$$\begin{aligned} \ln(1+a \cos x) \frac{1}{a \cos x} &= \int_0^1 \frac{dy}{1+ay \cos x}, \\ -\ln(1-a \cos x) \frac{1}{a \cos x} &= \int_0^1 \frac{dy}{1-ay \cos x}, \end{aligned}$$

$$\text{二式相加 } \ln \frac{1+a \cos x}{1-a \cos x} \frac{1}{a \cos x} = 2 \int_0^1 \frac{dy}{1-a^2 y^2 \cos^2 x},$$

所以 $I = 2a \int_0^{\frac{\pi}{2}} \left[\int_0^1 \frac{dy}{1-a^2 y^2 \cos^2 x} \right] dx$ 化为累次积分;

下面考虑交换积分次序: 注意 $|a| < 1$, 因而

$$\frac{1}{1-a^2 y^2 \cos^2 x} \text{ 在 } 0 \leq x \leq \frac{\pi}{2}, 0 \leq y \leq 1 \text{ 时连续,}$$

上面积分可以交换次序, 此后利用积分换元,

$$I = 2a \int_0^1 \left[\int_0^{\frac{\pi}{2}} \frac{dx}{1-a^2 y^2 \cos^2 x} \right] dy \stackrel{t=\tan x}{=} 2a \int_0^1 \left[\int_0^{+\infty} \frac{dt}{1-a^2 y^2 + t^2} \right] dy,$$

其中利用了 $dx = \frac{dt}{1+t^2}$ 以及 $\cos^2 x = \frac{1}{1+\tan^2 x} = \frac{1}{1+t^2}$;

继续积分便得

$$\begin{aligned} I &= 2a \int_0^1 \left(\frac{1}{\sqrt{1-a^2 y^2}} \arctan \frac{t}{\sqrt{1-a^2 y^2}} \bigg|_0^{+\infty} \right) dy \\ &= a\pi \int_0^1 \frac{dy}{\sqrt{1-a^2 y^2}} = \pi \arcsin a \text{。} \end{aligned} \quad \square$$